

Physics A (H156, H556)

Exploring - Ultrasound ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **45**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) Ultrasound B-scans can be used to image unborn babies.

The figure below shows a B-scan of an unborn baby.

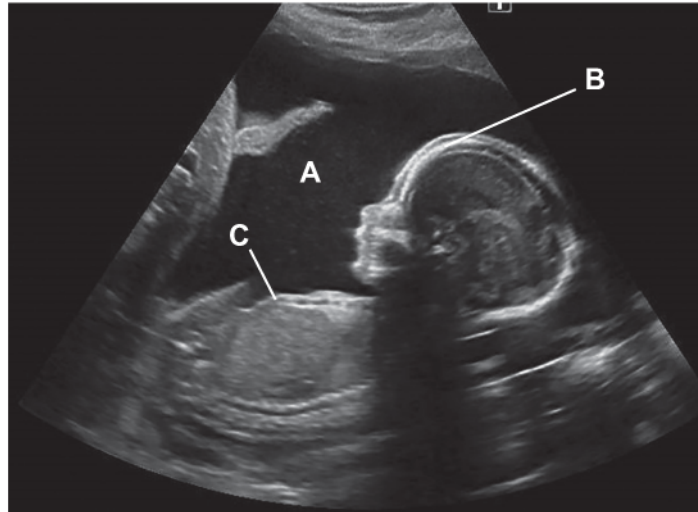


Fig. 17.2

i. Explain why no signal is received back from A.

----- [1]

ii. Explain why a greater signal is received back from B than C.

----- [2]

- (b) Doppler ultrasound can be used to measure the speed of blood flow through blood vessels.

The speed of ultrasound in blood is 1600 ms^{-1} .

A transducer emitting ultrasound of frequency 10.0000 MHz is placed at 50° to the blood vessel.

The reflected ultrasound has a frequency of 9.9987 MHz .

Calculate the speed v of the blood flow.

$$v = \dots\dots\dots \text{ms}^{-1} \quad [2]$$

- (c) Explain what is meant by ultrasound.

----- [2]

- (d) Fig. 17.1 is a labelled photograph of an ultrasound examination of a patient.

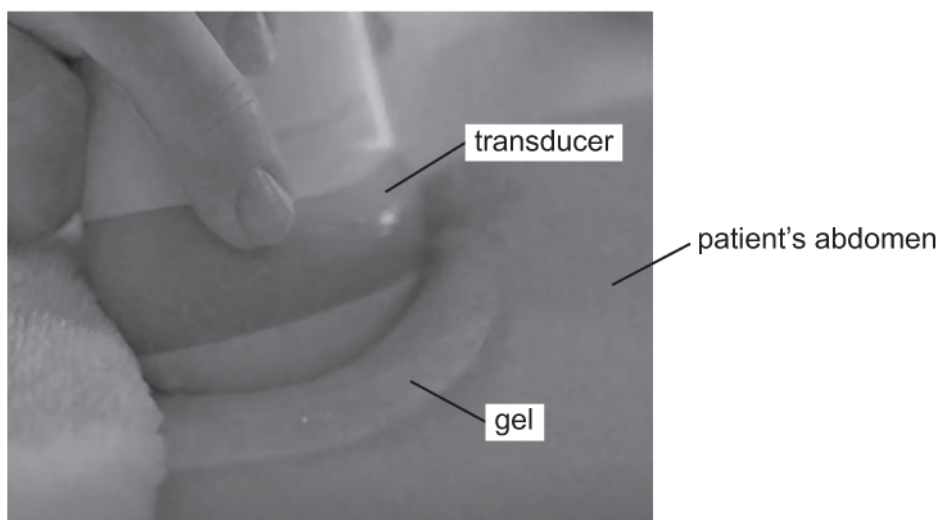


Fig. 17.1

Explain how the transducer both produces and receives ultrasound waves.

Explain the purpose of the gel.

[6]

2 A beam of ultrasound is incident normally at a boundary between two tissues F and G.

The table below shows some data on the two tissues.

	Tissue F	Tissue G
Density of tissue	ρ	1.2ρ
Speed of ultrasound in tissue	c	$1.5c$

What percentage of the intensity of the ultrasound is reflected at the boundary?

- A 0.83%
- B 8.2%
- C 9.1%
- D 29%

Your answer

[1]

3(a) This question is about the medical use of ultrasound.

There are several different types of ultrasound scanning techniques.

Explain how an A-scan could be used to measure the thickness of a patient's eye lens.

You may draw a diagram to help with your answer.

[3]

(b) In ultrasound scanning, explain what is meant by **impedance (acoustic) matching** and how it may be achieved.

[2]

4 The Doppler effect is used to measure the speed of blood flow in arteries.

Which medical diagnostic method uses this technique?

- A barium scan
- B CAT scan
- C PET scan
- D ultrasound scan

Your answer

[1]

5(a) Explain how an ultrasound transducer can **emit** ultrasound.
You do **not** need to describe the design of the transducer.

[2]

(b) Explain how the reflection of ultrasound at a boundary between two tissues depends on the physical properties of the tissues.

[3]

6 A contrast material is used while taking an X-ray image of a patient.
Which statement is correct?

- A Iodine is a contrast material.
- B Technetium is a contrast material.
- C A contrast material must have a short half-life.
- D A contrast material is used for acoustic matching.

Your answer

[1]

7 The acoustic impedance Z of a material in the shape of a cube can be determined using the equation

$$Z = \frac{Mc}{L^3}$$

where M is the mass of the material, L is the length of each side of the cube and c is the speed of ultrasound in the material.

The percentage uncertainty in L is 1.2 % and the percentage uncertainty in c is 1.8 %. The percentage uncertainty in M is negligible.

What is the percentage uncertainty in Z ?

- A 2.2 %
- B 3.0 %
- C 4.2 %
- D 5.4 %

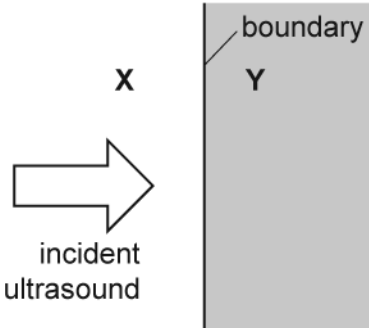
Your answer

[1]

8 The table shows some data on two tissues in a patient.

Tissue	Density	Acoustic impedance
X	ρ	$1.5 Z$
Y	1.3ρ	Z

Ultrasound in tissue X is incident at the boundary between the tissues X and Y.



What is the percentage of the ultrasound intensity reflected at the boundary?

- A 1.7 %
- B 4.0 %
- C 13 %
- D 20 %

Your answer

[1]

9(a) A patient with a blood clot in his muscle is having an ultrasound A-scan.
Fig. 24.1 shows an ultrasound transducer placed on the patient's skin.

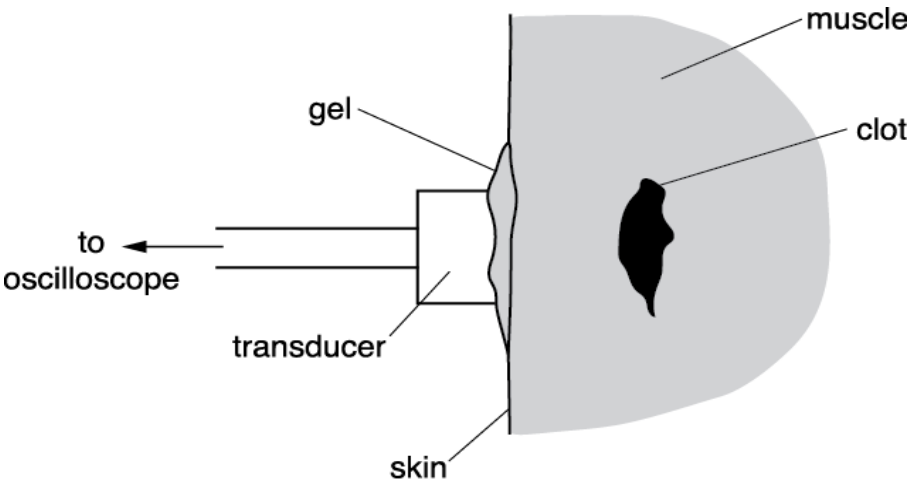


Fig. 24.1 (not to scale)

The ultrasound transducer produces pulses of ultrasound. An oscilloscope is connected to the transducer. Fig. 24.2 shows part of the oscilloscope display.

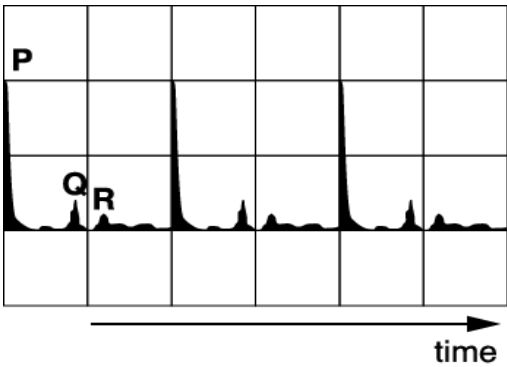


Fig. 24.2

In Fig. 24.2, P is one of the signal pulses produced by the transducer.
Explain the origin of the pulses Q and R.

[2]

- (b) The **front** of the blood clot is 1.5 cm from the skin.
The density of the patient's muscle is 1070 kg m^{-3} .
The time difference between pulses **P** and **Q** in Fig. 24.2 is $19 \text{ }\mu\text{s}$.
Determine the acoustic impedance Z of patient's muscle. State an appropriate unit for your answer.

$Z =$ _____ unit: _____ [4]

- 10 Ultrasound is reflected at a boundary between two materials.

Which property of the materials governs the intensity of the ultrasound reflected at the boundary?

- A density
- B decay constant
- C acoustic impedance
- D attenuation coefficient

Your answer

[1]

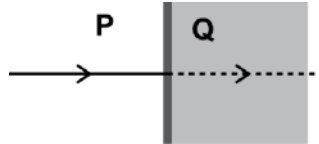
- 11 What is the correct SI unit for acoustic impedance?

- A kg s
- B $\text{kg m}^{-2} \text{ s}^{-1}$
- C $\text{kg m}^{-3} \text{ s}^{-1}$
- D $\text{kg m}^{-2} \text{ s}^{-2}$

Your answer

[1]

- 12 The diagram below shows a beam of ultrasound incident at the boundary between two materials.
The acoustic impedance of material P is 1.5 times the acoustic impedance of material Q.



What is the percentage of the ultrasound intensity **transmitted** at the boundary?

- A 20 %
- B 30 %
- C 80 %
- D 96 %

Your answer

[1]

- 13(a) State what is meant by the *piezoelectric effect*.

[1]

- (b) In an experiment, a beam of ultrasound is directed at the boundary between two materials **A** and **B**. Fig. 24.1 shows the beam incident at right angles to the boundary between these two materials.

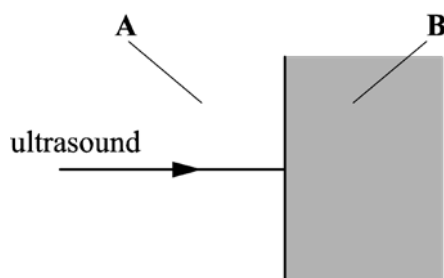


Fig. 24.1

The material **A** is unchanged. The acoustic impedance of material **A** is $2.5 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$. The material **B** is varied. The acoustic impedance of **B** is Z .

Fig. 24.2 shows the variation with Z of the percentage of reflected intensity of the ultrasound at the boundary.

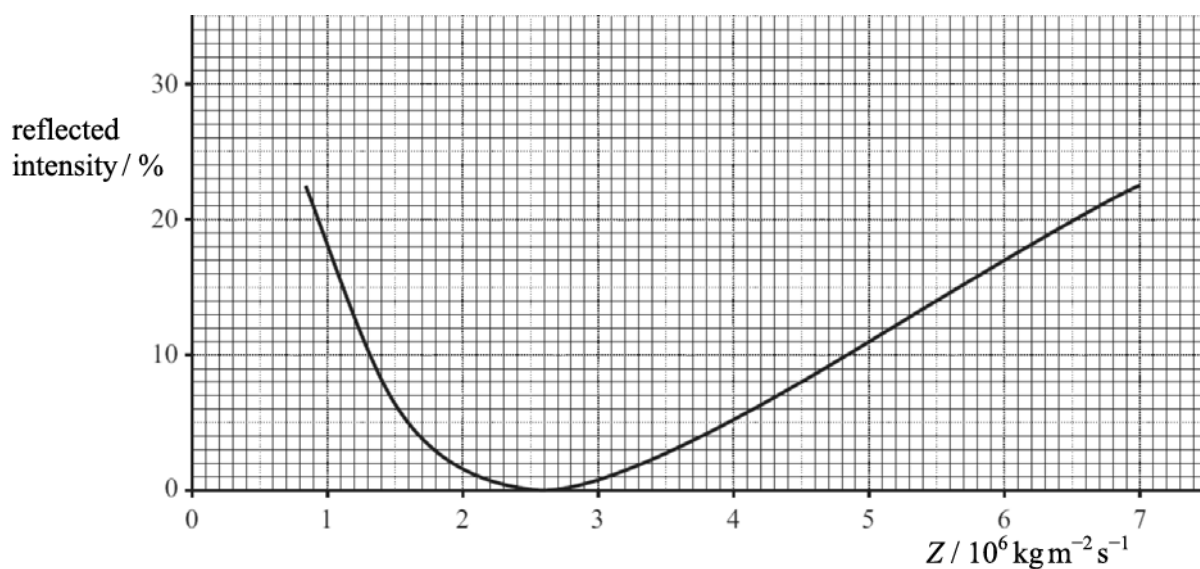


Fig. 24.2

Explain why the curve shown in Fig. 24.2 has a dip.

[3]

- [4]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a	i	There is no change in (acoustic) impedance	B1	ALLOW there is no change in the density (of the medium) ALLOW there is no boundary / there is nothing to reflect from NOT JUST there is nothing there IGNORE undefined symbols <u>Examiner's Comments</u> Many candidates gave vague answers such as 'it's all the same' or 'it is all air'. This response did require the idea that a boundary or some change must be present for the reflection to occur. Several candidates thought that it was absorbed at A. Around half of the candidates were able to score a mark on this question.
		ii	<u>Acoustic</u> impedance of C is different from B Change in <u>acoustic</u> impedance at B is greater (than at C) (relative to their surrounding medium/A)	C1 A1	IGNORE explanations in terms of density Second point gains both marks ALLOW soft tissue/skin/belly for C ALLOW skull/bone/head for B <u>Examiner's Comments</u> This question required the use of the term 'acoustic impedance' for any marks as this would lead to crediting quality responses. The term 'attenuation coefficient' was used by a large number of candidates in place of the acoustic impedance. Only around 40% used the term correctly and only half of this appreciated that this was the difference between B and C and their surrounding mediums rather than simply the difference between B and C. This question highlights the need for correct terminology and not vague statements.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$v = \Delta f c / 2 f \cos \theta$ $= \frac{(10.0000 \times 10^6 - 9.9987 \times 10^6) \times 1600}{2 \times 10.0000 \times 10^6 \times \cos 50}$ 0.16 (ms⁻¹)	C1 A1	IGNORE cancelling powers of 10 Correct to at least 2 significant figures <u>Examiner's Comments</u> This question was correctly done by around two thirds of candidates. Although not a particularly complex calculation, there are some challenges in it. Common mistakes were using the speed of light for <i>c</i>, using the reflected frequency for the change of frequency, power of ten errors and using the reflected frequency in the denominator. Many candidates wrote out the formula and rearranged it for <i>v</i> before substituting the numbers which help with the calculation. Those who avoided this occasionally made substitution/transcription errors.
	c	Longitudinal waves With a frequency greater than 20 kHz	B1 B1	ALLOW 20 kHz and above <u>Examiner's Comments</u> Many candidates attempted to answer this question in terms of the use of ultrasound, presumably prompted by the first line of the question. While many good responses were given along these lines, it was not the purpose of the question. Many candidates gave vague answers such as 'outside of the human hearing range' but a little under a half of candidates knew (and wrote) the value of 20 kHz. Only around one fifth were able to state it was a longitudinal wave.
	d	Level 3 (5–6 marks) Clear explanation of transducer and clear explanation of gel <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear explanation of transducer OR clear explanation of gel	6 x B1	Use level of response annotations in RM Assessor Indicative scientific points may include: Transducer • High frequency alternating PD applied to the faces of the piezoelectric crystal. • Causes the piezoelectric crystal to stretch and compress / oscillate / resonate at high frequency. • Ultrasound waves sent in pulses

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	or Some explanation of transducer and some explanation of gel <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited explanation of transducer or gel <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark <i>No response or no response worthy of credit</i>		- Between pulses, reflected ultrasound incident on the piezoelectric crystal causes the crystal to change shape / oscillate / resonate - An alternating PD is induced Gel - Acoustic impedance of air is different to body - Acoustic impedance of gel is similar to body - This allows greater transmission of the ultrasound waves into / out of the patient's body - There is less reflection of the ultrasound waves at the body boundary - Reference to $\frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$ <u>Examiner's Comments</u> This level of response (LoR) question was designed to assess Section 6.5.3 (b), (e) and (f) of the specification. A holistic approach to marking is used, with marks given according to answers matching the descriptors for the various levels. No one answer is perfect for this question and examiners will expect a varied approach. The indicative scientific points are used as a basis; however other correct, relevant physics may also be taken into account. The nature of this question is such that it can be conveniently separated into two sections; a description of producing and receiving ultrasound by the transducer, and the purpose of the gel in this case. The key points in the production and receiving of ultrasound that examiners were looking for were: - an explanation of the piezoelectric effect - what causes it, and the effect on the crystal - how this produces the ultrasound waves

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					• how a reverse of this will lead to its detection. The key points in the purpose of the gel were:- an explanation of acoustic impedance and an idea of what would happen if no gel were to be used – some idea of the term impedance matching and how this occurs with the gel. It was clear that many candidates had a good knowledge and understanding the use of ultrasound in this context and were able to describe this well, leading to a good number of candidates being given Level 3. The very best answers were detailed and well-structured and made every attempt to fully answer each section of the question using correct scientific terms. Many candidates wasted time and space describing how the ultrasound was reflected in the body, which is not required, or other irrelevant detail. Some drew diagrams, which can be helpful, but in this case did not really add to the text. For the most part, the description of the gel was done better than the description of production, most likely because the piezoelectric effect can be difficult to explain clearly. Candidates were not penalised for the use of incorrect values in the frequency of ultrasound here, or in any values of size (or voltage applied to) the transducer. The equation for reflection of ultrasound was given by several candidates but was not always explained to support the answer. Many strong candidates only answered one part of the question; this highlights the need to read it carefully – there will nearly always be two parts – and make sure that, as far as possible, equal weighting is given to both.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2			B	1	<u>Examiner's Comments</u> The correct response is B. The formula booklet is pretty essential here as the ultrasound reflection formula is one of the least memorable. Although there are a couple of stages to this calculation, most candidates were able to answer this correctly and this suggests a confidence (and perhaps suitable practice) in this topic. However, and unusually, some candidates did not provide a response to this question.  OCR support An awareness of the Data, Formulae and Relationship booklet and its content and structure is valuable here, avoiding the need to recall potentially complex formulae. Candidates can save themselves time by using the booklet frequently in preparation for the examination and thereby gain an appreciation of its value.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a		(Pulses of) ultrasound sent into the eye	B1	Allow ultrasound reflected by any part of the eye Allow 'sound' / wave (since ultrasound is in the question) Ignore transducer placed close / next to eye
			Reflections from front and back of lens (and pulses displayed on oscilloscope)	B1	
			(Thickness of lens) determined from speed (of ultrasound) and time (difference)	B1	Allow thickness $= \frac{ct}{2}$ with c = speed (of ultrasound) and t = time (difference) Allow this mark even when the reflections are from incorrect boundaries
	b		(special coupling) gel is used that has the same / 'matching' (acoustic) impedance as skin / body	B1	Allow Z Allow gel and impedance is the same / matching for two materials / mediums
			Reduced / less / zero reflection (at the skin)	B1	Allow more transmission
			Total	5	
4			D	1	
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
5	a	Piezoelectric (effect) mentioned An alternating p.d. applied to the film / crystal produces vibrations / resonance (and this in turn produces ultrasound)	B1 B1	Allow 'applied p.d. changes the shape of a crystal / film' Allow alternating current / a.c. Allow expand and contract for vibrations (AW) Allow a named film / crystal Examiner's Comments Many candidates showed good knowledge of the piezoelectric effect. A significant number of candidates incorrectly believed that the vibrations of the piezoelectric film, or the crystal, were because of a steady potential difference applied to its end; a very high frequency alternating e.m.f. is required to force the crystal to oscillate. The modal score for this question was 2 marks.
	b	density (of tissue) or speed (of ultrasound in tissue) or acoustic impedance mentioned Same Z, no reflection / different Z gives reflection $I_r/I_0 = [Z_1 - Z_2]^2/[Z_1 + Z_2]^2$ and intensity mentioned	B1 B1 B1	Not ρ or c or Z Allow same Z, total transmission / different Z gives some transmission Allow fraction of intensity reflected = $(Z_1 - Z_2)^2/(Z_1 + Z_2)^2$ Examiner's Comments The majority of the candidates scored a mark for mentioning acoustic impedance of the tissues. This term was confused with attenuation coefficient (from X-rays) by some of the candidates. Examiners could not give any credit if this confusion was evident in the answers. A pleasing number of candidates knew that the reflection of ultrasound at the boundary was significant when the difference in the acoustic impedances was large, or the equivalent, that there was no reflection when the acoustic impedances matched. For many candidates, it was the difference in density of the tissues, and not acoustic impedance, that governed the fraction of the reflected intensity at the boundary. Only a small percentage of the candidates omitted this question.
		Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6			A	1	<u>Examiner's Comments</u> The majority of the candidates did get the correct answer A. A significant number of candidates opted for C, confusing contrast material with medical tracers.
			Total	1	
7			D	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8			B	1	
			Total	1	
9	a		The reflection of the ultrasound produces the pulses Q and R	B1	
			Pulses Q and R are due to reflections from the front and back of the clot	B1	
	b		$\text{speed} = 2 \times 1.5 \times 10^{-2} / 19 \times 10^{-6} (= 1579 \text{ m s}^{-1})$ $Z = \rho c = 1070 \times 1579$ $Z = 1.7 \times 10^6$ unit: $\text{kg m}^{-2} \text{ s}^{-1}$	C1 C1 A1 B1	Allow 2 marks for 8.4×10^5 ; factor of 2 omitted
			Total	6	
10			C	1	
			Total	1	
11			B	1	
			Total	1	
12			D	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
13	a		The material expands or contracts when a p.d. is applied across its opposite faces.	B1	Allow: When a p.d. is applied across its opposite faces the material expands or contracts.
	b		The fraction f of the incident intensity of ultrasound reflected at the boundary is given $f = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$ There is reflection when $Z \neq 2.5 \times 10^6 \text{ (kg m}^{-2} \text{ s}^{-1}\text{)}$ At $Z = 2.5 \times 10^6 \text{ (kg m}^{-2} \text{ s}^{-1}\text{)}$ there is impedance (acoustic) matching and hence no reflection of ultrasound.	B1 B1 B1	
	c		The transducer is placed at an angle to the arm or artery and ultrasound is reflected by the moving blood cells. The wavelength or the frequency of the reflected ultrasound is altered. Since $\Delta f = \frac{2vf\cos\theta}{c}$, the change in frequency $\propto$ speed of the blood. The technique is non-invasive / no incision needed / minimises risk of infection.	B1 B1 B1 B1	
			Total	8	